Variable-Thrust, Variable-Mass Rockets

From orbital trajectory planning to predictive control

Applied Mathematical Methods | 2026

01 Problem & physical model

Orbit insertion requires both altitude and velocity. Near-Earth orbits enable observation, space experiments and subsequent missions. Reaching altitude without sufficient transverse speed yields a suborbital return.

Vertical ascent and ideal propulsion

For expelled mass $-\mathrm{d}m > 0$ and constant effective exhaust speed c, momentum balance gives

$$m\,\mathrm{d}v = -c\,\mathrm{d}m + F_{\mathrm{ext}}\,\mathrm{d}t.$$

Neglecting external forces and integrating yields the Tsiolkovsky equation:

$$\boxed{\Delta v_{\mathrm{ideal}} = c\ln(m_0/m_f)}.$$

Impulse is $I_T = \int T\,\mathrm{d}t = c(m_0 - m_f)$; velocity gain integrates T/m. Starting from rest with $r \geq r_0$, no random forcing and dissipative drag, ideal propulsion and orbital energy imply

$$\frac{m_0}{m_f} \geq \exp\left[\frac{\sqrt{2\mu/r_0 - \mu/r_*}}{c}\right].$$

The necessary mass ratio is at least **6.245** in the 600 km case; this bound is not sufficient for insertion.

Earth-centered planar polar dynamics

Define $u = \dot{r}$, $w = r\dot{\theta}$, $V = \sqrt{u^2 + w^2}$, propellant flow $q = -\dot{m} \geq 0$ and thrust $T = cq$.

$$\dot{r} = u, \qquad \dot{\theta} = w/r, \qquad \dot{m} = -q,$$
$$\dot{u} = \frac{w^2}{r} + \frac{cq\cos\alpha}{m} - \frac{\mu}{r^2} - \frac{kVu}{m} + a_{\varepsilon,r},$$
$$\dot{w} = -\frac{uw}{r} + \frac{cq\sin\alpha}{m} - \frac{kVw}{m} + a_{\varepsilon,\theta}.$$

Here $k = \rho C_D A/2$ and $\rho = \rho_0 e^{-(r-R_E)/H}$. Thrust follows $\alpha = \mathrm{atan2}(w,u) + \delta$ after a short initial vertical hold.

$$a_{\varepsilon,j} = \sigma_{0,j}(\rho/\rho_{\mathrm{ref}})\eta_j(t).$$

Gaussian forcing is continuously interpolated with finite temporal correlation, not ideal white noise. The planar point-mass model omits Earth rotation, lift and attitude actuator dynamics.

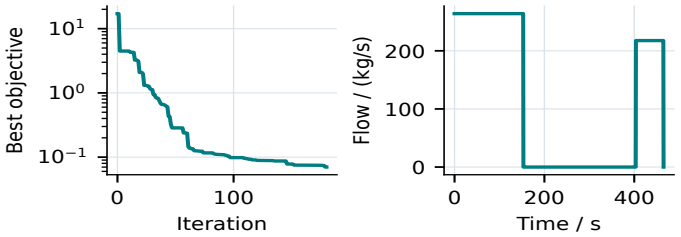
02 Planning & feedback control

Two burns: PSO search + SQP refinement

A suborbital burn, coast and circularization burn structure the search over thrust-angle nodes, flow rates and phase durations. PSO combines personal and neighborhood experience; SQP then refines terminal feasibility.

$$r_* = R_E + 600\,\mathrm{km}, \qquad v_c = \sqrt{\mu/r_*},$$

$$J = \left(\frac{r_f - r_*}{s_h}\right)^2 + \left(\frac{u_f}{s_u}\right)^2 + \left(\frac{w_f - v_c}{s_w}\right)^2.$$



Left: PSO objective history. Right: final reference flow. Burn/coast/burn durations are approximately 153.3 / 250.0 / 62.0 s. Final accuracy reflects both PSO and local refinement.

Forward integration

Runge–Kutta integration maps controls to trajectories. Batched GPU RK4 accelerates search; adaptive Dormand–Prince integration checks candidates with fuel-depletion and ground-impact events.

MPC: measure, predict, solve, apply

Track $x^{\mathrm{ref}} = (r, \theta, u, w, m)^{\mathsf{T}}$ using thrust-angle and flow corrections. Linearizing the discrete dynamics along the reference gives

$$e_{k+1} = A_k e_k + B_k v_k + d_k.$$

Quadratic penalties on predicted tracking errors, control effort, control changes and slack variables are condensed into a convex quadratic program:

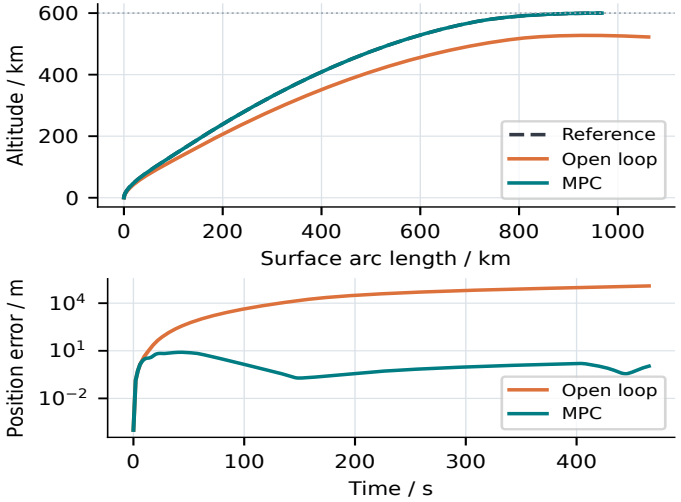
$$\min_{\zeta}\ \tfrac{1}{2}\zeta^{\mathsf{T}} H\zeta + f^{\mathsf{T}}\zeta, \qquad G\zeta \leq b.$$

A **convex interior-point algorithm** solves the QP with angle, flow, dry-mass and ground constraints. Apply only the first correction, then repeat with fresh state feedback; retain the two-burn schedule.

Offline planning supplies a feasible reference; online MPC corrects disturbances in flight.

03 600 km case & findings

$m_0 = 60\,000$ kg, $m_{\mathrm{dry}} = 1\,200$ kg, $c = 4\,500$ m/s. Open and closed loops share the same plant disturbance, distinct from the planning realization.



MPC nearly overlaps the reference trajectory. Position error combines radial and surface-arc errors; zero values are floored only for logarithmic plotting.

Metric	Open loop	MPC
Position RMS / m	5.98×10^4	2.606
Velocity RMS / (m/s)	298.084	0.0992
Cutoff altitude error / m	−77 611.6	1.191
Cutoff radial speed / (m/s)	−557.983	0.0231
Tangential speed error / (m/s)	233.256	0.0559

RMS is relative to the reference; cutoff errors are relative to the target circular orbit.

Feedback suppresses error accumulation and brings cutoff close to circular-orbit conditions.

Scope: one disturbance realization, ideal state feedback and a simplified atmosphere. No claim of global optimality, statistical robustness or proven closed-loop stability is made.

References: NASA Glenn, Ideal Rocket Equation; Kennedy & Eberhart, Particle Swarm Optimization, 1995; Dormand & Prince, JCAM, 1980; Mayne et al., Automatica, 2000; Nocedal & Wright, Numerical Optimization, 2006. Data: project experiment snapshot, 20 Oct 2026.